Chapter test

11 Transpiration, transport and support in plants

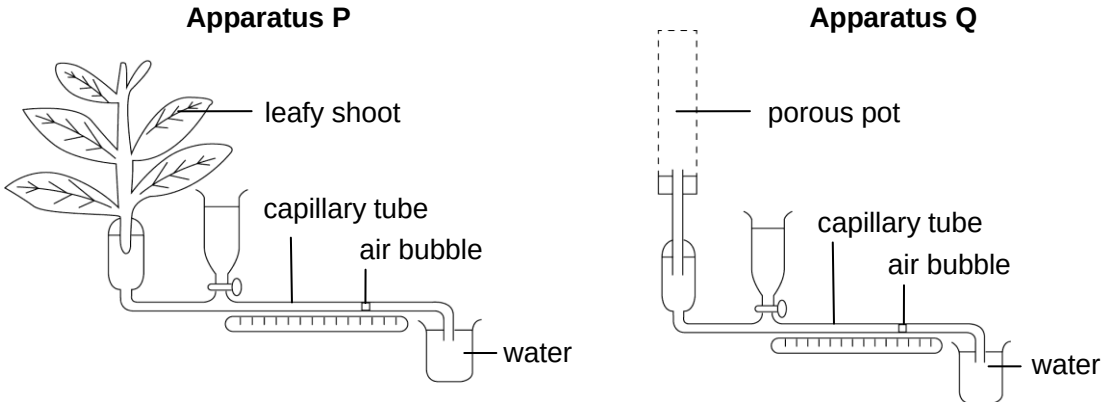
Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

***Directions*:** Questions 1 and 2 refer to the experimental set-up below.



1. The leafy shoot is cut and connected to the apparatus under water because this can

(1) allow water to enter the xylem vessels more easily.
(2) prevent air from entering the xylem vessels.
(3) prevent plasmolysis of the cells of xylem.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

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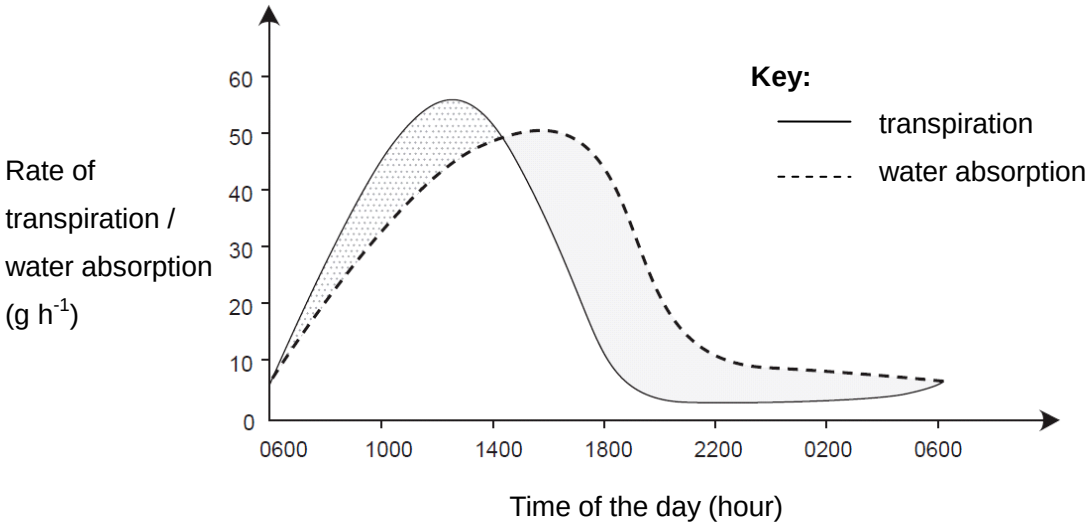
★ 2. Which of the following correctly shows how the distance travelled by the air bubble changes when apparatus P and Q are moved from a well-lit place to a dark environment?

	Distance travelled by the air bubble in apparatus P	*Distance travelled by the air bubble in apparatus Q*
A.	decreases	decreases
B.	decreases	remains unchanged
C.	increases	remains unchanged
D.	increases	increases

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***Directions*:** Questions 3 and 4 refer to the graph below, which shows the rates of transpiration and water absorption of a plant for 24 hours.



★ 3. The increase in transpiration rate from 0600 to 1300 was probably due to

A. an increase in the rate of photosynthesis.
B. an increase in the rate of water absorption.
C. an increase in air temperature.
D. an increase in the rate of respiration. ☐

★ 4. At what time does the plant show the sign of wilting?

A. 0600
B. 1200
C. 1500
D. 1800 ☐

★ 5. Which of the following combinations of environmental conditions would most likely increase the transpiration rate?

	Relative humidity	***Wind speed***	***Air temperature***
A.	low	low	low
B.	high	low	high
C.	low	high	high
D.	high	high	low

☐

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★ 6. Which of the following features allow terrestrial plants to reduce loss of water?

(1) The leaves are broad and flat.
(2) The stomata close when light intensity is low.
(3) The upper epidermis of leaves has fewer stomata than the lower epidermis.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

7. Which of the following processes in plants involve(s) active transport?

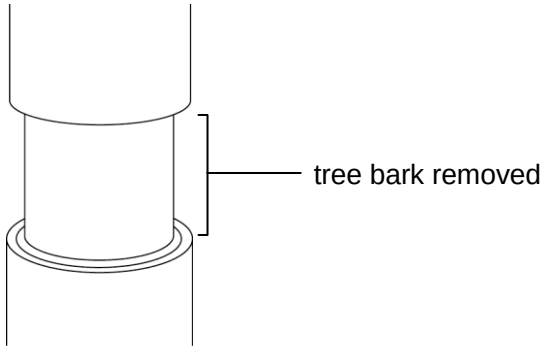
(1) transpiration
(2) absorption of minerals from the soil
(3) transport of water in the stem

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only ☐

★ 8. Which of the following could help us distinguish xylem from phloem?

A. The cells of xylem have cell walls.
B. The cells of phloem carry out respiration.
C. The cells of xylem are located in the vascular bundles.
D. The cells of phloem are not present in the roots. ☐

★ 9. A strip of bark including the phloem has been removed from the stem of a tree.

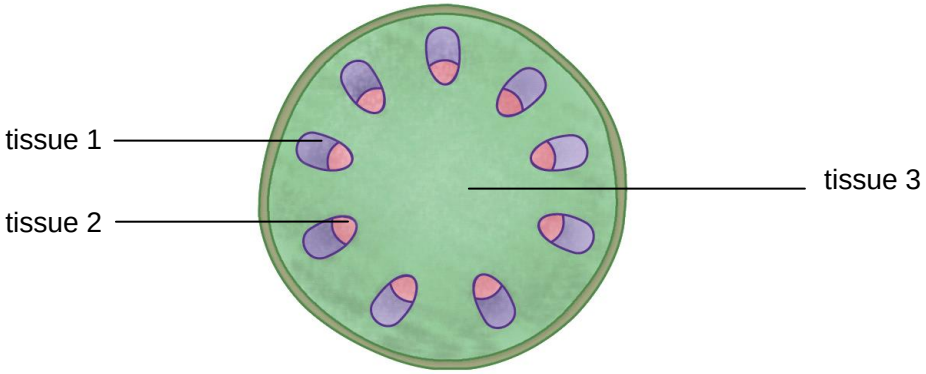


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Which of the following is the possible consequence?

A. Amino acids cannot be transported to the roots.
B. Water cannot be transported to the leaves.
C. Water cannot be transported to the roots.
D. Minerals cannot be transported to the leaves. ☐

★★10. The diagram below shows the cross section of a stem.



Which of the following statements about the water potential of the cells in the three tissues is correct?

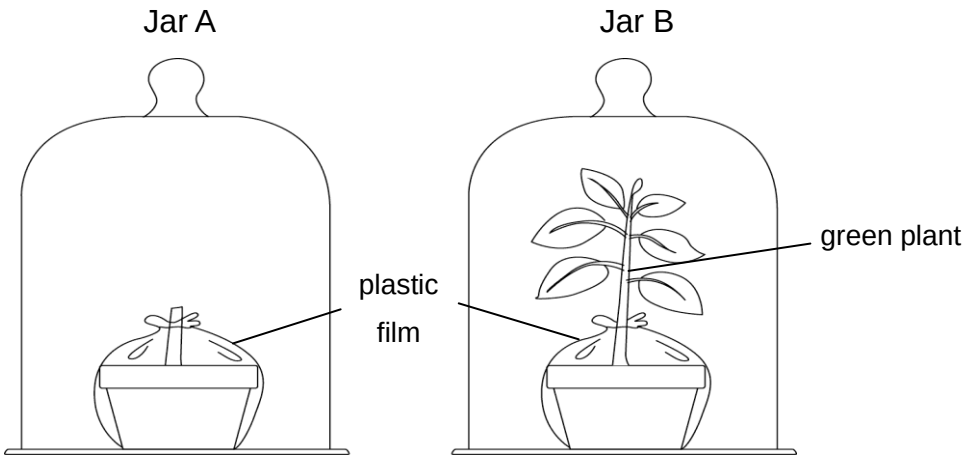
A. The water potential of tissue 3 is higher than that of tissue 2.
B. The water potential of tissue 2 is higher than that of tissue 1.
C. Tissues 2 and 3 have the same water potential.
D. Tissues 1, 2 and 3 have the same water potential. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	11.1	11.1	11.1	11.1	11.1	11.1	11.2	11.2	11.2	11.2

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B. Problem solving (16 marks)

11. The diagram below shows the set-up of an experiment.



After 24 hours, there were some droplets on the inner wall of jar B while there were no droplets on the inner wall of jar A.

(a) What is the purpose of

(i) covering the soil with a plastic film? (1 mark)

(ii) setting up jar A? (1 mark)

(b) Why did droplets appear on the inner wall of jar B? (2 marks)

(c) Suggest ***one*** method that can be used to prove that the droplets are water. (2 marks)

(Total: 6 marks)

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★12. A student carried out a study on the number of stomata in the leaves of two plants (A and B). Plant A grows in an area under strong sunlight. Plant B grows in a shaded area. The table below shows the findings.

	Average number of stomata on the lower epidermis (mm^2)
Plant A	89
Plant B	46

(a) Guard cells control the opening and closing of stoma. State ***two*** structural differences between guard cells and epidermal cells. (2 marks)

(b) Suggest and explain ***one*** advantage of having a larger number of stomata on the lower epidermis in plant A. (2 marks)

(c) Which plant, A or B, would you expect to have a thicker cuticle in its leaves? (1 mark)

(Total: 5 marks)

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★ 13. On a hot summer day, the stem of a herbaceous plant becomes soft and the plant cannot stand upright.

(a) In the space below, draw a labelled diagram to show the structure of a palisade mesophyll cell when the plant becomes wilted. (3 marks)

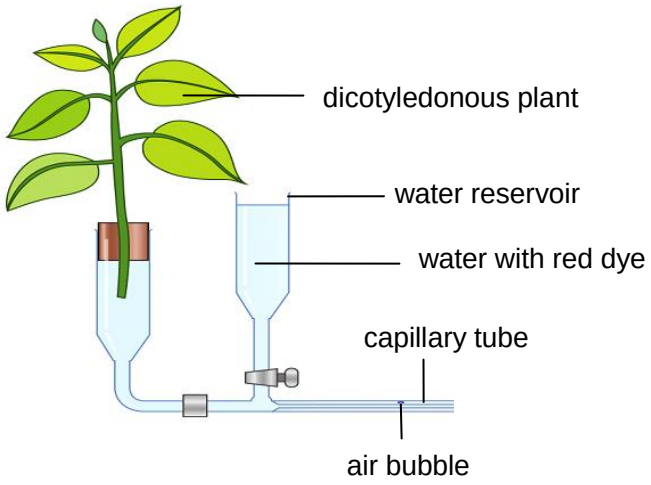
(b) On the same day, the stems of the woody plants remain upright. Explain this with reference to the support mechanism of woody plants. (2 marks)

(Total: 5 marks)

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C. Exam challenge (14 marks)

★★14. The diagram below shows a part of an apparatus used to investigate the rate of transpiration in a dicotyledonous plant. The apparatus was put on an electronic balance, and then placed in a well-illuminated and well-ventilated room for five hours. The water in the apparatus had been pre-mixed with a red dye.



(a) (i) Name the apparatus. (1 mark)

(ii) State the function of the water reservoir. (1 mark)

(iii) State ***two*** precautions that should be taken for accurate reading. (2 marks)

(b) The initial and final positions of the air bubble in the capillary tube were recorded. The initial and final readings in the electronic balance were also taken. The results are shown below.

	Initial reading	**Final reading**
Positions of the air bubble in the capillary tube (cm^3)	50.0	48.5
Electronic balance (g)	168.2	167.1

(Given: the density of water is 1 g/cm^3)

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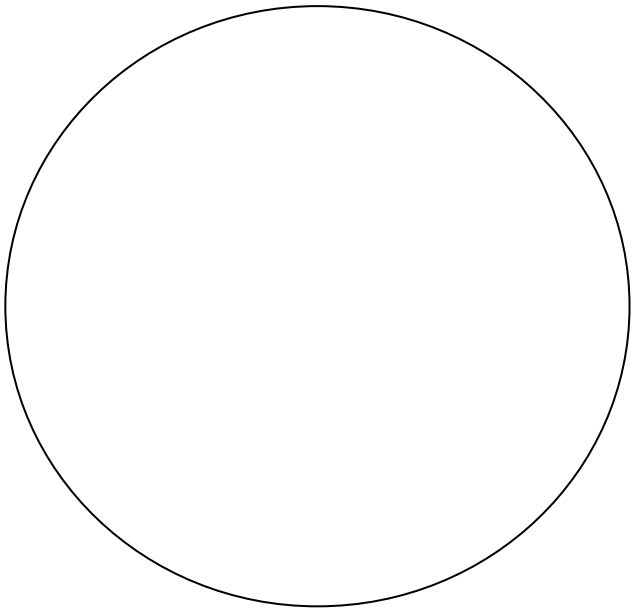
(i) Calculate the change in the readings of the capillary tube and electronic balance respectively. (2 marks)

(ii) What does the change in the reading of the capillary tube represent? (1 mark)

(iii) What does the change in the reading of the electronic balance represent? (1 mark)

(iv) What are the significances of the difference between the change in the reading of electronic balance and the capillary tube to the growth of plant? (3 marks)

(c) A slide of stem was cut transversely and observed under a microscope. Draw a labelled low-power diagram to show the distribution of vascular bundles in the stem section and shade the parts that are stained red. (3 marks)



(Total: 14 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	B	2.	B	3.	C	4.	B	5.	C
6.	C	7.	B	8.	B	9.	A	10.	A

(1 mark each)

B. Problem solving (16 marks)

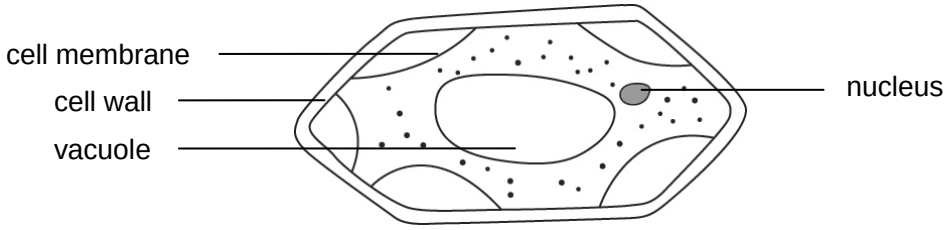
11. (a) (i) To prevent water loss / evaporation of water from the soil. (1)
(ii) To show that the droplets formed in jar B come from the plant only. (1)
(b) Water vapour is lost from the leaves by evaporation. (1)
The water vapour condenses on the inner wall of the jar. (1)
(c) Use a piece of anhydrous cobalt chloride paper to test the droplets. (1)
If the droplets are water, the paper will change from blue to pink. (1)

(Total: 6 marks)

12. (a) Guard cells have chloroplasts but epidermal cells do not. (1)
The inner cell walls of guard cells are thicker than the outer cell walls. The inner and outer cell walls of epidermal cells usually have an even thickness. (1)
(b) This can increase the rate of transpiration / water loss (1)
because transpiration can produce a cooling effect to prevent overheating under strong sunlight. (1)
(c) Plant A (1)

(Total: 5 marks)

13. (a) Correct drawing: Cell membrane separated from the cell wall (1)
Correct labelling (1)
Correct title (1)



The structure of a palisade mesophyll cell of a wilted plant

(b) The stems of woody plants have a large proportion of xylem / thick-walled cells. (1)
Therefore, they are mainly supported by the rigidity / mechanical or physical strength of xylem / thick-walled cells. (1)

(Total: 5 marks)

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C. Exam challenge (14 marks)

14. (a) (i) Potometer (1)

(ii) When the tap is open, the water inside the water reservoir will push the air bubble back to the original position. (1)

(iii) Any ***two*** of the following: (2)

- Cut the shoot under water. (1)
- Insert the shoot into the potometer under water. (1)
- Dry the leaves before measuring the weight of the set-up. (1)
- Smear the junctions with vaseline / ensure no water leakage appears in the set-up. (1)
- Allow time for the plant to reach equilibrium before taking measurements. (1)

(b) (i) Change in reading of the capillary tube = 50.0 – 48.5 = 1.5 cm^3 (1)

Change in reading of the electronic balance = 168.2 – 167.1 = 1.1 g (1)

(ii) The amount of water absorbed by the plant (1)

(iii) The amount of water lost / transpired by the plant (1)

(iv) The change in the reading of the capillary tube is larger than that of the electronic balance. (1)

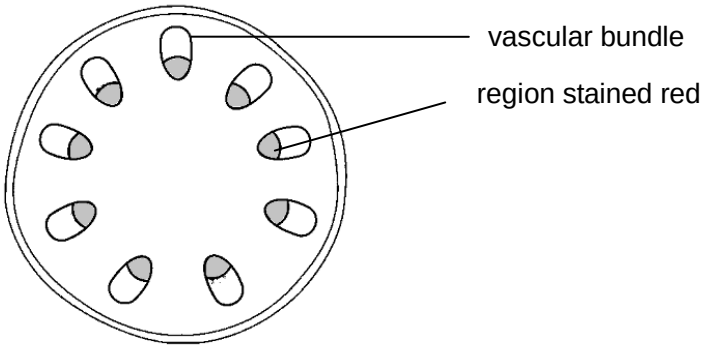
This shows that there is a net gain of water by the plant. (1)

The water gained is used for photosynthesis / supporting the plant / formation of new cells, etc. (1)

(c) Correct drawing (1)

Correct labelling (1)

Correct shading (1)



(Total: 14 marks)

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